

Category C Assignment Problems (Any Five)

P1. Voting Eligibility Checker

Scenario

SRM's student council election desk needs a one-line check at the registration table: is this student old enough to vote in the campus election?

Task

- Accept a student's age as an integer.
- Build a boolean expression $\text{age} \geq 18$.
- Using a single if / else statement, print "Eligible to vote" when the expression is true, otherwise print "Not eligible to vote".
- No loop is needed for this one — just a boolean expression and if / else.

Suggested Method Signature(s)

```
void checkVotingEligibility(int age)
```

Sample Input / Output

Input	Output
age = 20	Eligible to vote
age = 16	Not eligible to vote

Concepts covered: Boolean expressions, comparison operators, if / else, the boolean data type.

P2. Positive, Negative, or Zero Classifier

Scenario

A basic calculator utility needs to label any number a student types in as positive, negative, or exactly zero, before running further calculations on it.

Task

- Accept an integer number.
- Using an if / else if / else ladder, check whether the number is greater than 0, less than 0, or equal to 0.
- Print "Positive", "Negative", or "Zero" accordingly.

Suggested Method Signature(s)

```
void classifyNumber(int number)
```

Sample Input / Output

Input	Output
number = 15	Positive
number = -4	Negative
number = 0	Zero

Concepts covered: if / else if / else, relational operators, three-way decision-making.

P3. Print Numbers 1 to N

Scenario

A warm-up drill for new coding-club members: print a simple countup so beginners can see a for loop execute step by step.

Task

- Accept a positive integer N.
- Use a for loop that starts at 1 and runs while the counter is less than or equal to N.
- Print each number on its own line.

Suggested Method Signature(s)

```
void printNumbersUpToN(int n)
```

Sample Input / Output

Input	Output
n = 5	1 2 3 4 5

Concepts covered: for loops, loop counters, loop conditions, basic iteration.

P4. Sum of Natural Numbers Using a While Loop

Scenario

A mental-math practice tool asks a student for a number N and instantly shows the sum of every natural number from 1 up to N, to check their own manual addition.

Task

- Accept a positive integer N.
- Use a while loop with a counter starting at 1, adding the counter to a running total on each pass, and incrementing the counter until it exceeds N.
- Print the final sum.

Suggested Method Signature(s)

```
void sumOfNaturalNumbers(int n)
```

Sample Input / Output

Input	Output
n = 5	Sum of numbers from 1 to 5 = 15

Concepts covered: while loops, accumulator variables, loop counters, incrementing.

P5. Day Name From Number

Scenario

A campus timetable app stores each weekday as a number (1 = Monday ... 7 = Sunday) internally, but needs to display the actual day name to students.

Task

- Accept an integer from 1 to 7.

- Use a switch statement with a case for each number, printing the matching day name.
- Include a default case that prints "Invalid day number" for anything outside 1–7.
- Remember to include break after each case so execution doesn't fall through into the next one.

Suggested Method Signature(s)

```
void printDayName(int dayNumber)
```

Sample Input / Output

Input	Output
dayNumber = 3	Wednesday
dayNumber = 9	Invalid day number

Concepts covered: switch statement, case labels, break, the default case.

LEVEL 2 PROBLEMS

5 problems · two concepts combined · logical operators with conditionals, loops with break/continue, switch with validation

P1. Grade Classifier With Logical Operators

Scenario

The STEP progress sheet needs a grade band that also checks attendance eligibility together with the marks — a student only gets a real letter grade if they also cleared the minimum 75% attendance bar; otherwise they're marked "Detained" regardless of marks.

Task

- Accept a student's marks (0–100) and attendance percentage (0–100).
- First check, using a boolean expression with the && operator, whether attendance is 75 or above AND marks are 40 or above.
- If either condition fails, print "Detained".
- Otherwise, using an if / else if / else ladder on marks, print the grade: 90+ -> "A", 75–89 -> "B", 60–74 -> "C", 40–59 -> "D".

Suggested Method Signature(s)

```
void classifyWithAttendance(int marks, int attendance)
```

Sample Input / Output

Input	Output
marks = 82, attendance = 80	Grade: B
marks = 91, attendance = 60	Detained

Concepts covered: Logical AND (&&), boolean expressions, if / else if / else, combining multiple conditions.

P2. Login Attempt Simulator With Break

Scenario

A simple login screen for a coding-club internal tool gives a user up to 3 attempts to enter the correct access code before locking them out.

Task

- Accept the correct access code and an array of up to 3 attempted codes.
- Use a for loop over the attempts, checking each one against the correct code.
- The moment a correct attempt is found, print "Access granted on attempt X" and use break to exit the loop immediately.
- If the loop finishes without a match, print "Access denied — all attempts used".

Suggested Method Signature(s)

```
void simulateLogin(String correctCode, String[] attempts)
```

Sample Input / Output

Input	Output
correctCode = "1234", attempts = {"0000", "1234", "9999"}	Access granted on attempt 2
correctCode = "1234", attempts = {"1111", "2222", "3333"}	Access denied — all attempts used

Concepts covered: for loops, if statements, break, early-exit logic.

P3. Skip Multiples Printer Using Continue

Scenario

A numbers-practice worksheet wants every number from 1 to 20 printed, except multiples of 3, which should be silently skipped rather than shown as zero or blank.

Task

- Use a for loop from 1 to 20.
- Inside the loop, check whether the current number is a multiple of 3 using the modulus operator.
- If it is a multiple of 3, use continue to skip printing it and move straight to the next iteration.
- Otherwise, print the number.

Suggested Method Signature(s)

```
void printSkippingMultiplesOfThree()
```

Sample Input / Output

Input	Output
(loop from 1 to 20)	1 2 4 5 7 8 10 11 13 14 16 17 19 20

Concepts covered: for loops, the modulus operator, continue, conditional skipping.

P4. Menu-Driven Calculator Using Switch

Scenario

A quick calculator demo for orientation day reads two numbers and an operator choice, then performs the matching arithmetic using a menu built from a switch statement.

Task

- Accept two double values and a single character representing the operator: '+', '-', '*', or '/'.
- Use a switch statement on the operator character, with a case for each of the four operators, printing the result.
- For '/', check whether the second number is 0 before dividing, printing "Cannot divide by zero" instead if so.
- Include a default case that prints "Invalid operator" for anything else.

Suggested Method Signature(s)

```
void calculate(double a, double b, char operator)
```

Sample Input / Output

Input	Output
a = 10, b = 4, operator = '+'	Result: 14.0
a = 10, b = 0, operator = '/'	Cannot divide by zero
a = 10, b = 4, operator = '%'	Invalid operator

Concepts covered: switch statement, case labels, default case, combining switch with if for extra validation.

P5. Prime Number Checker

Scenario

A number-theory mini-quiz for the coding club needs a quick way to check whether a given number is prime, showing the check stopping the instant a factor is found rather than needlessly continuing.

Task

- Accept an integer greater than 1.
- Use a boolean flag initialized to true, and a for loop checking divisibility from 2 up to number / 2.
- The moment any divisor is found, set the flag to false and use break to stop checking further.
- After the loop, print "Prime" or "Not Prime" based on the flag.

Suggested Method Signature(s)

```
void checkPrime(int number)
```

Sample Input / Output

Input	Output
number = 17	Prime
number = 18	Not Prime

Concepts covered: for loops, boolean flags, break, the modulus operator, early-exit optimization.

LEVEL 3 PROBLEMS

5 problems · several concepts together · nested loops, loops with boolean flags and break, switch inside loops, full decision-and-repetition logic

P1. Number Pyramid Pattern Printer

Scenario

A classroom whiteboard exercise on nested loops asks students to reproduce a simple numeric pyramid, one row taller than the last, to build comfort with loops inside loops.

Task

- Accept the number of rows N.
- Use an outer for loop for each row from 1 to N.

- Use an inner for loop that prints the row number that many times on the same line.
- Move to a new line after each row completes.

Suggested Method Signature(s)

```
void printNumberPyramid(int n)
```

Sample Input / Output

Input	Output
n = 4	1 2 2 3 3 3 4 4 4 4

Concepts covered: Nested for loops, outer/inner loop counters, pattern printing.

P2. ATM PIN Retry System

Scenario

A campus mess-card ATM kiosk gives a student up to 3 tries to enter the correct 4-digit PIN before the card is blocked, and must stop the moment the PIN is entered correctly.

Task

- Accept the correct PIN and a list of up to 3 attempted PINs.
- Use a while loop with an attempt counter starting at 0 and a boolean flag for success, continuing while attempts remain and the flag is still false.
- Inside the loop, check the current attempt against the correct PIN using a boolean expression; if it matches, set the flag to true and print "PIN accepted", then break.
- If the loop ends without success, print "Card blocked — too many incorrect attempts".

Suggested Method Signature(s)

```
void atmPinRetry(String correctPin, String[] attempts)
```

Sample Input / Output

Input	Output
correctPin = "4821", attempts = {"1111", "4821"}	PIN accepted
correctPin = "4821", attempts = {"1111", "2222", "3333"}	Card blocked — too many incorrect attempts

Concepts covered: while loops, boolean flags, break, boolean expressions, combining loop and conditional logic.

P3. Traffic Signal Simulator

Scenario

A road-safety awareness demo for orientation day cycles a traffic signal through Red, Green, and Yellow for a fixed number of cycles, using a switch statement to decide what each state means.

Task

- Accept the number of cycles to simulate.
- Use a for loop over the cycles, and inside it a switch statement on the current state index (0 = Red, 1 = Green, 2 = Yellow) to print the color and its meaning ("Stop", "Go", "Prepare to stop").
- After each switch, advance to the next state, wrapping back to Red after Yellow using the modulus operator.

Suggested Method Signature(s)

```
void simulateTrafficSignal(int cycles)
```

Sample Input / Output

Input	Output
cycles = 3	Red - Stop Green - Go Yellow - Prepare to stop

Concepts covered: switch statement, for loops, the modulus operator for wraparound, state cycling.

P4. Multiplication Table Generator With Input Validation

Scenario

A revision worksheet generator should refuse to print a multiplication table for an invalid number (0 or negative), reprompting conceptually before generating the table for the first valid number in a batch.

Task

- Accept an array of candidate numbers to try, in order.
- Use a for loop over the candidates; for each one, check with an if statement whether it is 1 or greater.
- If a candidate is invalid (less than 1), print "Skipping invalid number: ..." and use continue to move to the next candidate.
- The moment a valid candidate is found, print its multiplication table from 1 to 10 using a nested for loop, then use break to stop processing further candidates.

Suggested Method Signature(s)

```
void generateFirstValidTable(int[] candidates)
```

Sample Input / Output

Input	Output
candidates = {-3, 0, 7, 9}	Skipping invalid number: -3 Skipping invalid number: 0 7 x 1 = 7 7 x 2 = 14 ... 7 x 10 = 70

Concepts covered: for loops, continue, break, if validation, nested loops.

P5. Guess the Number Game

Scenario

A lightweight orientation-day game: the system has already picked a secret number, and the player keeps guessing until they get it right or run out of a fixed number of tries, getting a "too high" / "too low" hint after each wrong guess.

Task

- Accept the secret number, a maximum number of tries, and an array of the player's guesses in order.
- Use a while loop with a try counter, continuing while tries remain and the number hasn't been guessed yet.
- For each guess, use an if / else if / else to print "Too high", "Too low", or "Correct! You guessed it" and, on a correct guess, break out of the loop immediately.
- If the tries run out without a correct guess, print "Out of tries — the number was ...".

Suggested Method Signature(s)

```
void guessTheNumber(int secretNumber, int maxTries, int[] guesses)
```

Sample Input / Output

Input	Output
secretNumber = 42, maxTries = 4, guesses = {20, 60, 42}	Too low Too high Correct! You guessed it
secretNumber = 42, maxTries = 2, guesses = {10, 15}	Too low Too low Out of tries — the number was 42

Concepts covered: while loops, if / else if / else, break, boolean-driven loop conditions, combining every control-flow concept together.